AGENT SKILL · ENGLISH GUIDE

Academic Humanizer

Remove the AI tone from academic writing. Keep every number, equation, citation, and theorem.

v1.1 · October 2026

Academic Humanizer removes stock phrases, invented contrasts, cue colons, meta-narrative, defensive hedging, figurative and coined labels, and dashes from academic prose, rewrites methods and proofs that read as work logs, and states every claim at the strength its evidence or proven result supports. Numbers, equations, citation keys, theorem statements, and qualifiers that express real uncertainty stay as the author wrote them. A bundled Python checker verifies each of these invariants and lists every changed sentence. The skill runs in Claude, OpenAI Codex, ChatGPT, and any agent that reads SKILL.md.

github.com/junbolian/Academic-Humanizer · MIT License

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1

Overview

Academic Humanizer is an agent skill for editing and drafting manuscripts, theses, referee responses, and grant proposals in English and Chinese. It runs in Claude, OpenAI Codex, ChatGPT, and any agent that reads `SKILL.md`. A bundled checker verifies every edit, so the science in the text stays the science you wrote.

What it removes

- **Stock phrases and AI vocabulary.** delve, pivotal, sheds light on, “In recent years, X has attracted increasing attention”, “has important theoretical and practical significance”, and their Chinese counterparts.
- **Sentence patterns typical of model output.** Invented contrasts (“The barrier is not technology but understanding”), cue colons (“The key insight is:”), restatement sentences (“This means that”), adjacent sentences built on one skeleton, rhetorical lists, and ordinal headings.
- **Meta-narrative and defensive writing.** “In this section, we …”, “To summarize, …”, self-deprecation (“only a preliminary attempt”), and stacked hedges (“may potentially suggest”).
- **Figurative language and dashes.** Persona metaphors (“acts as a tireless reviewer”), figurative labels, and dashes used as punctuation.
- **Coined labels.** An abstract modifier plus a system noun that the text never defines or cites, such as “certification protocol”, “verification layer”, or “safeguard mechanism”. Each becomes what the text says it does. When the text never says, the report suggests what the author should state. Established terms such as “attention mechanism” stay.
- **Methods and proofs written as work logs.** “Verification pipeline”, “sanity check”, “audit”, “certified bound”, solver tolerances, random seeds, guards that never triggered, and script names. Computer-assisted proof steps become lemmas that say whether the computation is exact or numerical. Every value a reported result depends on stays, and every deleted number is listed in the report.
- **Claims that miss their evidence.** “Significantly” without a test, “optimal” where the theorem proves consistency, a vague “large gain” where the paper reports 3.1%, and hedges on results the paper itself proves.

What it keeps

Numbers, equations, citation keys, `\ref` and `\label` keys, the text of definitions, assumptions, and theorems, the LaTeX preamble, and every qualifier that expresses real uncertainty. Edits add no names, numbers, quotations, sources, or examples. A claim that needs evidence the text lacks receives a visible `[[EVIDENCE: ...]]` placeholder for the author to fill. While any placeholder remains, the report opens with a warning that the text is not ready to submit, and the checker ends its output with a `NOT READY TO SUBMIT` line.

Results that the evidence supports read as results. The voice the skill produces is that of a confident scholar: specific, assertive, and exact about scope.

Three modes

Mode	Use it for	What changes
Polish	drafts	Wording, flow, idiomatic English, and claim scope
Cleanup	final or submitted text	Only sentences that match a listed pattern, and only the matched part
Draft	new sections, letters, proposals	New text written under the same rules, with every reference searched and confirmed

Install

Claude Code

```
git clone https://github.com/junbolian/Academic-Humanizer.git
cp -r Academic-Humanizer/academic-humanizer ~/.claude/skills/
```

Claude apps. Zip the `academic-humanizer` folder and upload it as a custom skill.

OpenAI Codex. Copy the folder to `~/.agents/skills/` for all projects, or to `.agents/skills/` inside one repository. Invoke it with `$academic-humanizer`, or let Codex select it from the task.

ChatGPT. On Business, Enterprise, and Edu plans, upload the zipped folder under Plugins, Skills, Create, Upload from your computer. On other plans, create a GPT or a project, paste `platforms/chatgpt-instructions.md` into its instructions, attach the files of the `academic-humanizer` folder, and turn on code execution.

Any other agent. Load `SKILL.md` as instructions and give the agent access to `references/` and `scripts/`.

Use

Plain requests trigger the skill.

- "Polish the introduction in `paper.tex`."
- "This abstract is final. Remove the AI tone only."
- "Draft a reply to Reviewer 2, comment 3. We added Section 5.2."

Requests written in Chinese work the same way.

The agent returns the full revised text and a short report: changes grouped by rule, claims rescoped, placeholders to fill, structural notes, and the checker's verdict.

The checker

```
python3 academic-humanizer/scripts/check_revision.py audit paper.tex
python3 academic-humanizer/scripts/check_revision.py compare paper.tex revised.tex --changes
```

`audit` lists candidate phrases and patterns by line. `compare` exits with status 1 when any number, citation key, label, math segment, formal statement, or the preamble changes. It also lists every qualifier that disappeared, every new capitalized term and quotation, and every changed sentence, so each edit maps to a rule. The script needs Python 3 and nothing else.

Chapter 5 shows a complete checker run.

2

The skill: SKILL.md

The skill makes AI-assisted academic prose read the way a careful author in the field writes: literal, precise, connected from one sentence to the next, confident where the evidence is strong, and never stronger than the evidence. It changes wording, not content. Numbers, equations, citation keys, cross-references, formal statements, and genuine qualifiers stay as the author wrote them, and a bundled script verifies this. Authors keep their venue's AI-disclosure obligations.

Modes and scope

- **Polish** is the default when the user asks to polish, tighten, rewrite, 润色, or 改写. All Rules apply, and sentences may be rewritten for flow (Rule 8) and idiom (Rule 9) within the edit discipline below.
- **Cleanup** applies when the user asks only to remove AI tone, says 只去AI味 or 成稿清理, or calls the text final. Only Rules 1 to 4 and the first item of Rule 5 apply. A sentence that matches none of them stays verbatim, and a sentence that matches changes only in the matched part. Claims that need evidence go to the report, not into placeholders.
- **Draft** applies when the assistant writes new manuscript text, such as a section, a response letter, or a proposal. Apply the Rules while writing, run `check_revision.py audit` on the draft, and fix what it finds before delivering. Never cite from memory: search for every new reference, confirm that it exists, and check that the sentence describing it matches the source. A literature review includes a search for work from the last two years.

When the request does not settle the choice between polish and cleanup, use polish for drafts and cleanup for text the user calls final or submitted.

Edit discipline

1. **Every change has a rule.** Each change corresponds to a numbered Rule below. A change without a rule is reverted. Taste in wording is not a rule.
2. **Smallest sufficient change.** Within a sentence that matches a rule, change what the rule targets and leave the neighbouring words alone. Polish mode also allows Rules 8 and 9 to act on the same sentence.
3. **Nothing added.** Names, institutions, roles, places, numbers, dates, orderings, quotations, sources, links, causal claims, motives, and examples enter the text only if the original or the user's material contains them. Every content word in the revision traces to one of those two sources. Concrete detail is the author's to supply, and a thin passage becomes a report item.
4. **Nothing lost.** Claims, conclusions, concessions, and qualifiers that express real uncertainty stay: may, likely, typically, in some cases, under these conditions, 可能, 通常, 在某些情况下. Rule 7 removes two kinds of qualifier only, hedges on results that the text itself proves or reports and stacked duplicates, and the report lists each removal. Rule 11 removes engineering detail that no result depends on, and the report lists each removal with its numbers. `compare` lists every qualifier and number that disappears.

5. **Fixed structure.** Heading levels, section order, paragraph count and order, lists, tables, figures, quotations, and code stay in place unless the user asks for structural change. Structural problems go into the report (Rule 12).

Style reference first

When the user supplies the author's earlier papers, a style guide, or a target-voice description, read it before editing. Note sentence length, connectives, level of hedging, how sections and paragraphs open, punctuation habits, and notation. Where the reference conflicts with Rules 1, 2, 3, 4, 8, or 9, follow the reference, since those rules set register and the reference shows the author's register. An author who uses dashes on purpose keeps them. The reference never overrides Rules 5, 6, 7, and 10, which protect evidence, proven scope, stance, and the invariants.

Running the checker

SKILL_DIR is the folder that contains this file. Common locations are `~/.claude/skills/academic-humanizer` in Claude Code, `/mnt/skills/user/academic-humanizer` in the Claude apps, and `~/.agents/skills/academic-humanizer` or `.agents/skills/academic-humanizer` in OpenAI Codex. An environment without Python still gets the same comparisons: carry them out by reading, and state in the report that the script did not run.

Process for polish and cleanup

1. Save the original verbatim in a working folder, for example `ah/orig.tex`, or `.md` and `.txt` for other text. Record the document type, venue, language, and mode.
2. Read the style reference, if one was supplied.
3. Run `python3 SKILL_DIR/scripts/check_revision.py audit ah/orig.tex`. The output lists candidate AI-tell phrases with their lines, coined labels, sentence patterns, dashes, semicolon and parenthesis density, long sentences, runs of very short sentences, and the section and result structure. Each hit is a candidate for a decision. "Robust" is correct in a paper on robust statistics, and "navigation" is literal in a robotics paper.
4. Read the full text before changing anything. Identify each paragraph's single point and how it follows from the paragraph before, since flow problems show at this level.
5. Rewrite under the Rules and the edit discipline. Cleanup mode touches only sentences with a rule match.
6. Save the revision next to the original and run `python3 SKILL_DIR/scripts/check_revision.py compare ah/orig.tex ah/revised.tex --changes`. The script compares numbers, citation keys, `\ref` and `\label` keys, math, the text of formal environments, and the preamble. It then lists lost qualifiers, new capitalized terms, new quoted strings, and every changed sentence, and audits the revision against the original. Revert each unintended difference, match each changed sentence to a rule, and repeat until only intended differences remain.
7. Go through the final checklist and deliver as described under Output.

For a single paragraph under about 150 words the script is optional. Compare numbers, citations, qualifiers, and math against the original by reading before delivering.

Read `references/examples.md` before the first rewrite in a conversation. It shows each rule applied with and without the evidence a sentence needs, and the same paragraph in polish and cleanup modes.

Rules

1. Remove stock phrases and AI vocabulary

The full candidate list is `references/lexicon.txt`, which the script reads. The main groups:

- Vocabulary: delve, underscore, intricate, tapestry, testament, pivotal, showcase, foster, realm, seamless, harness, multifaceted, nuanced, holistic, meticulous, paramount, invaluable, crucial when it only means important, leverage when it only means use.
- Significance claims without content: paves the way, a crucial step toward, opens new avenues, sheds light on, bridges the gap, of paramount importance, transformative, groundbreaking, new paradigm.
- Formulaic openers: “In recent years, X has attracted increasing attention”, “With the rapid development of”, “Despite recent advances”, “X plays a crucial role in”. Open with the specific problem instead.
- Spoken openers: Simply put, Long story short, Bottom line, 说白了, 说穿了, 先说结论. Delete them and state the point.
- Size words without numbers: extensive or comprehensive experiments, a wide range of, numerous, various, substantial. Replace each with the count or the number the paper reports.
- Novelty padding: “novel” more than once per section, “for the first time”, and “to the best of our knowledge”. Keep at most one, and only where the authors searched the literature.
- Announced emphasis: It is worth noting that, Importantly, Interestingly, Notably. State the fact and let its position and evidence carry its weight.
- Copula avoidance (“serves as” or “stands as” where “is” is meant), negative parallelism for emphasis (“not just X but Y”), trailing “-ing” clauses that add a vague evaluation (“…, highlighting the importance of …”), and lists of three where one item is meant.
- Implications boilerplate: rich insights, important implications for practitioners, provides a new perspective, delicate balance, interplay, repeated counterintuitive.
- Response-letter boilerplate: thanks for an “insightful comment” in every reply. Thank the reviewers once, at the top of the letter. Each reply states what changed and where.

2. Write literally

Replace metaphors and figurative labels with the mechanism they stand for: landscape, through the lens of, cornerstone, blueprint, roadmap, double-edged sword, sweet spot, unlock, arms race, holy grail, silver bullet, at the heart of, tackle, grapple with. A reader in the field checks each sentence against the model or the data, and a metaphor offers nothing to check.

Persona metaphors are the clearest case. A tool, model, or procedure described as an idealized person (“acts as a tireless reviewer”, “like a wise mentor”, 像一位智慧的导师) is usually followed by praise or a “not only … but also” clause. State what the thing does: “it checks each constraint against the data in seconds”.

Established technical terms stay even when they began as metaphors: bottleneck, halo effect, spillover, curse of dimensionality, cold start, warm start, signal and noise, backbone for a network architecture,

loss landscape, heuristic as the name of an algorithm class. A term stays when it has a definition in the field or in the paper. A precise description beats a general label: “a method without a performance guarantee” says more than “a heuristic method”.

The rule exists for checkability. Metaphor as such does not separate model output from human writing, and human writers of Chinese nonfiction use more of it (see “Not reasons to change”). Apply it to papers, theses, reports, and letters. In essays and popular writing where figurative language belongs to the genre, follow the style reference.

The paper’s title and the names the authors gave to models, policies, and effects stay. A figurative name gets one mention in the report, and the decision stays with the author.

Coined labels. Generated text often names an ordinary step with an abstract modifier and a system noun: “certification protocol”, “verification layer”, “safeguard mechanism”, “trust framework”, “evaluation pipeline”, 认证协议, 保障机制, 治理闭环. The label reads like a component the reader should already know, yet the text never defines it, cites a source for it, or uses it again as a component, and it often stands for a single check. The script’s “Coined labels” section lists each such phrase with its count and with whether the text defines, cross-references, or cites it. Decide each one:

- A term the field defines (attention mechanism, clinical trial protocol, incentive mechanism, hidden layer), or a term the paper defines and reuses, stays. A name the authors gave their own method also stays and gets one mention in the report when it says less than the method does.
- When the text states what the label does, replace the label with that statement. “Each solution then passes a certification protocol” becomes “We then check each solution for feasibility and recompute its objective with the exact solver” when both steps appear elsewhere in the text.
- When the text never states what the label does, keep the label and add a report suggestion: name the steps it consists of, or use the plain noun for what happens (“the check”, “the second review”). Never supply the steps yourself.
- A modifier that asserts a property, such as certified, verified, guaranteed, safe, or robust, is a claim. Check it under Rules 5 and 6: “certification” promises a guarantee that a proof or a stated procedure must back.
- A label that renames an object the text already names goes back to that name (Rule 8). A revision never introduces a label, and `compare` lists every label that is absent from the original.

3. Punctuation: no dashes, few semicolons and parentheses

Remove em dashes (U+2014, LaTeX `---`, the Chinese 破折号) and spaced en dashes or hyphens used as dashes. Use a comma, a colon, or two sentences. En dashes in numeric ranges (`2--6`) and in joint names (`Karush--Kuhn--Tucker`) stay, and so do Chinese 连接号 as described in `references/chinese.md`.

Semicolons and parenthetical asides follow the same logic. Two or more semicolons in one paragraph, or a parenthetical aside in most sentences, signal a paragraph that needs restructuring into full sentences. Parentheses around citations, equation numbers, units, and abbreviations at first use are standard and stay.

Dash rates vary by model by more than an order of magnitude. In a corpus of Chinese nonfiction, measured per 1,000 characters, human writers used 0.80 dashes, Claude 4.25, DeepSeek 5.16, Kimi 2.32, Gemini 0.51, and GPT 0.11. The same corpus shows GPT using 不是……而是 most often of the five models. Every draft gets the full check whichever model produced it.

4. Sentence patterns that mark generated text

Each pattern below appears at a clearly higher rate in model output than in human writing, measured on 300 model-written articles from five models and 329 human-written articles in Chinese nonfiction, and each has a trigger that can be located in a specific sentence. The English forms make the same move in English. An item applies only when its trigger is present.

4.1 Invented contrasts. The sentence sets up a misconception nobody in the text holds and then corrects it: "The real barrier is not technology but understanding", "X is not about A; it is about B", "What matters is not A but B", 不是……而是, 并非……而是, 不在于……而在于, 与其说……不如说, 看似……实则, 你以为……其实, 说到底. State the claim directly: "The barrier is understanding." A contrast stays when someone the text identifies holds A, such as a prior result, an assumption the paper relaxes, or a hypothesis the data reject: "Unlike \citet{ref1}, we do not assume a closed population." Rule 1 covers the additive form "not just X but Y".

4.2 Cue colons and list lead-ins. A cue phrase and a colon announce content instead of stating it: "The key insight is: …", "The answer is simple: …", "The result: …", "In short: …", 一句话总结：, 核心是：, 关键在于：, 原因如下：, 本质上：. Delete the cue when it carries nothing. When it links to the previous sentence, keep it and replace the colon with ordinary syntax. A sentence that only announces a list ("Here are the main reasons:", 我见过的几种典型场景：) becomes a statement with content or goes, and the list follows the preceding text with every item intact. Colons that introduce a quotation, a definition, an equation, or the parts of a general statement stay ("The model has two components: a measurement model and a structural model"), as do genre lead-ins such as "We make three contributions:" in many introductions and "The proof has two steps:". Headings, list-item labels, URLs, and code are out of scope.

4.3 Repeated sentence skeletons and filler examples. Two or more adjacent sentences built on one skeleton, with the same clause order, comma positions, and length, read like a completed form. The measured rate is 9.41 such pairs per 100 paragraphs in model text against 4.81 in human text. Change the syntax of one sentence by merging, splitting, reordering, or subordinating it, and keep its content. When a paragraph gives three or more parallel cases that make the same point, keep the most representative case in the main clause and compress the others into a clause without dropping information. Narrative sequence, enumerated steps, and parallel clauses inside one sentence stay.

4.4 Restatement sentences. A sentence that opens with "This means that", "This suggests that", "In other words", 这意味着, 这表明, 这说明, or 换句话说 and repeats the previous sentence merges into that sentence or keeps only its conclusion. A sentence that draws a conclusion the previous one does not contain stays, and its support is checked under Rules 5 and 6.

4.5 Rhetorical lists. One clause chaining three or more parallel items to sound complete, such as "startups, mid-sized firms, and large groups all benefit", becomes "firms of every size benefit". Without a fitting category word, change the syntax of one item. Lists that carry content stay complete: datasets, baselines, variables, assumptions, parameter values, regulatory items, and Markdown lists. Most lists in academic text carry content.

4.6 Ordinal headings. In reports, handouts, posts, and slides, three or more consecutive headings numbered 一、二、三, 第一、第二, or First, Second lose the ordinals and keep their text. Section numbers set by a manuscript class file or venue template, numbered steps, and items referred to by number are out of scope.

4.7 Paragraph openings without a referent. A paragraph other than the first opens with a comment such as 听起来, 看起来, 值得注意的是, 更重要的是, 关键在于, or 问题在于 and contains no word pointing back to the preceding text (这, 那, 其, 此, 上述). Add the referent: 这听起来像……. In English, a paragraph that opens with "More importantly," or "The catch is" names its subject.

4.8 Signpost connectives at paragraph starts. When 然而, 因此, 此外, 与此同时, 换言之, 总而言之, Moreover, Furthermore, However, or Therefore opens several consecutive paragraphs, keep the connective only where it marks the logical relation at a turn in the argument. Move an additive connective into the sentence or delete it. Rule 8 governs connectives inside paragraphs.

4.9 Meta-narrative, self-reference, previews, and recaps. Sentences about the text itself carry no content: "In this section, we discuss …", "We now turn to …", "As mentioned above, …", "Before proceeding, …", "To summarize, …", "This section has shown …", "The remainder of this paper is organized as follows", 本节将介绍, 下面我们将, 如前所述, 本文结构安排如下, 本章小结. Delete them, or keep only the substantive claim they wrap ("As shown above, the bound is tight" becomes "The bound is tight (Proposition 2)"). The paper's conclusion section keeps its summary. Cross-references that point to content the reader needs at that point stay: "the bound in Section 3 gives …".

5. Empirical claims match their evidence

- When a summary phrase ("substantially improves", "a large gain", 显著提升, 实现了效率的提升) stands where the same or the next paragraph gives the number, date, or object, put that specific value in its place. When the text gives no number, restore the plain verb ("achieves an improvement in accuracy" becomes "improves accuracy", 完成了对流程的优化 becomes 优化了流程) and add nothing.
- Each empirical claim points to its evidence, a table, figure, test, or citation, and its verb matches that evidence: "show" or "find" for what the reported results exhibit, "suggest" or "is consistent with" for indirect evidence. "Prove" is reserved for proofs.
- Replace a vague magnitude with the number the paper reports, with its metric, comparator, and sample: "reduces mean absolute error by 3.1% relative to the strongest baseline (Table 4)".
- Numbers, test results, table and figure references, and citations come only from the supplied material. When a sentence needs evidence that was not supplied, insert the visible placeholder `[[EVIDENCE: what is needed]]`, written in LaTeX as `\textbf{[[EVIDENCE: what is needed]]}` with plain words inside (no `%`, `_`, `&`, `#`). The script lists every placeholder and excludes its contents from the comparison. A placeholder is a correct output, and a plausible number is a fabrication. A text that still contains a placeholder is not ready to submit, and the report says so (see Output). Cleanup mode lists the need in the report instead.
- Report the statistic the paper computed, a mean with its standard error or confidence interval, a median, or a range, together with its aggregation. One statistic is never converted into another.
- Compare against the strongest baseline first.
- "Significant" and "significantly" require a stated test. Without one, give the difference and drop the word.
- All reported numbers come from one sample of record. Text that mixes samples, runs, or data versions without saying so is flagged in the report, and the numbers stay untouched.
- The phrasing states exactly what the evidence supports, neither weaker (Rule 7) nor stronger.

6. Theoretical claims match the proven statement

- For results proven in the paper, “we prove”, “we show”, and “we establish” are correct and stay.
- A sentence that restates a result carries its conditions: the assumptions, the class of methods, the parameter region, and whether the result is asymptotic. “The estimator is unbiased” becomes “Under Assumptions 1 and 2, the estimator is unbiased when the sampling fraction exceeds the threshold in Proposition 3.” Use the paper’s notation and names and introduce no new symbols.
- Check each quantifier against the statement: optimal, always, never, robust, for any. A result that depends on a functional form or a distribution, such as linearity or normal errors, says so wherever the introduction or the conclusion states it.
- Practical implications are conditional comparative statics with a direction and a source: “The number of sampling rounds needed for a given precision decreases in the capture probability and increases in the heterogeneity across strata (Proposition 4).” Advice such as “practitioners should carefully balance cost and precision” says less than the proposition it replaces.
- Theorems, propositions, and lemmas are never added, removed, merged, or promoted, except that Rule 11 states a computer-assisted proof step as a lemma. Suggestions about their number go into the report (Rule 12).

7. Stance: assert what the evidence supports

The target voice is a confident scholar: assertive verbs, specific results, no superlatives without evidence, and no apology.

- State proven and reported results as results. “Our method may reduce error by 3.1% (Table 4)” becomes “Our method reduces error by 3.1% (Table 4)”. Hedges stay only where uncertainty is real: indirect evidence, extrapolation beyond the sample, conjecture.
- Collapse stacked hedges to one: “may potentially”, “might possibly suggest”, “could perhaps”, “在一定程度上可能”.
- Remove self-deprecation: “this simple attempt”, “a preliminary step”, “our modest contribution”, “admittedly”, “we humbly”, 本文只是初步尝试, 抛砖引玉, 笔者水平有限, 不足之处敬请指正.
- Remove defensive writing: rebuttals of objections nobody raised, “we do not claim to …”, repeated “beyond the scope of this paper”, and weaknesses listed without bearing on a result’s validity.
- Remove unnecessary disclosure, such as accounts of failed attempts, of the order in which work was done, or of doubts the evidence does not raise. A limitation that bounds a result’s validity is stated once, plainly, where the result is stated or in the limitations paragraph. Disclosures that the venue requires, such as AI use, data availability, and conflicts of interest, stay where the venue puts them.
- State contributions directly: “We show”, “We introduce”, “This paper establishes”. Claims of being first, largest, or only stand when the evidence and a literature search support them.

8. Connected flow: one point per paragraph, no jumps

- Each paragraph makes one point, stated in its first or second sentence, and the remaining sentences support it.
- Each sentence follows from the one before. When the logical step between two sentences is missing, add a clause that states the relation already present in the argument: because, so, as a result, in contrast, for example, under this condition. When the relation is unclear, flag it in the report.

- Logical connectives stay: because, therefore, thus, however, in contrast, whereas. Additive connectives (Moreover, Furthermore, Additionally, In addition) at the start of consecutive sentences give way to the actual relation or to nothing.
- Split sentences longer than about 35 words, or with three or more subordinate clauses, so that each carries one step, and keep the pieces connected. Rhythm is not a reason to split (see "Not reasons to change").
- Runs of short declarative sentences, fragments such as "That limits progress.", and headline colons such as "The result: …" are errors too. A long sentence that states one condition and its consequence is fine.
- One object keeps one name. A defined term is reused exactly, and a repeated name is not replaced by a pronoun for variety. "Participants", "subjects", and "respondents" do not alternate for one group.
- Present tense for what the paper does and what its results show, past tense for procedures carried out.
- Definitions, assumptions, and result statements stay verbatim, and the compare script checks them.

9. Idiomatic English

Many drafts come from authors whose first language is not English, and the edit leaves the English idiomatic as well as free of AI phrases. Fix articles, countability ("a research", "informations"), prepositions (different from, consistent with, depend on), "which" versus "that", and constructions carried over from Chinese ("with the development of", "play an important role in", "make a contribution to", chains of "so as to"). Each replacement is a real collocation: "the methods remain largely empirical" is idiomatic, "the methods stay empirical" is not. Technical terms never become everyday words.

10. Preserve what is correct

- Qualifiers that express real uncertainty: suggests, is consistent with, may, we conjecture. "Suggests" does not become "shows" without new evidence. Rule 7 handles hedges on established results, and Rule 1 handles vague hedges such as somewhat and fairly.
- Passive voice when the actor does not matter, and "we" for the authors.
- Technical terms, notation, equations, LaTeX macros and environments, labels, and citation keys, verbatim.
- Genre conventions of careful authors: "Our work relates to three streams of literature" in an introduction, a contribution list, the structure of proofs.
- Citations: no citation key is added or removed. A cited work is described only from the existing text or from a source read in the current conversation, because descriptions written from memory are a frequent source of error. A citation list that never says how the works differ from the paper becomes a report item.

Not reasons to change

The features below look like AI writing to many readers, but corpus measurements show no gap, or a gap in the human direction. Changing them moves text away from human writing, so they never justify an edit. Rates are per 1,000 Chinese characters unless stated.

Feature	Measured result
Uneven sentence or paragraph length	No gap: the coefficient of variation of sentence length is 0.58 for models and 0.67 for humans.
Passive voice, nominalization, long sentences	Normal formal writing. Abstract passives such as 被认为 and 被视为 occur 0.09 times in model text.
Repeating a full name instead of a pronoun	More common in human text (0.04 against 0.02 adjacent-sentence repeats).
Parallel phrasing inside one sentence	As common or more common in human text, with ratios from 0.61 to 1.10 across forms.
Questions in body text, question headings	Body questions run 1.83 in human text against 0.10 in model text. Question headings show no gap as a share of all headings.
"First … second" inside paragraphs	No gap. Ordinal headings (4.6) are the target.
Few function words or colloquial connectives	Models use fewer, but added function words cannot be verified and push formal text toward speech.
Metaphor as such	Human writers use about 2.4 times as many.

11. Write methods and proofs as a paper, not as a work log

Generated methods sections and proofs often read as a record of running code: a "pipeline" of stages, a "verification protocol", a "sanity check", an "audit" of the results, a "certified bound", the tolerance a solver used, a guard that never fired. A paper states the method, the result, and the argument that establishes the result. Polish and draft modes apply this rule. Cleanup mode changes nothing under it and lists each item in the report.

- **Framing.** Methods and proofs do not use protocol, pipeline, workflow, sanity check, audit, certified bound, verification layer, or guardrail as framing. Name the object instead: the estimator, the algorithm, the bound in Proposition 2, Lemma 3. "A sanity check confirms that the bound holds" becomes the lemma or the computation that shows it holds. A bound is called certified only when a proof or an exact computation in the text establishes it, and then the text states it as a result with its proof. Terms the field defines stay, as in Rule 2: clinical trial protocol, cryptographic protocol, interval arithmetic.
- **Computer-assisted steps.** A proof step that rests on a computation is stated as a lemma and proved. The statement gives the inequality or identity, its domain, and its constants, all taken from the claim the text already makes. The proof says what was computed and by which method, and it says whether the computation is exact or numerical. Exact means symbolic, rational, or interval arithmetic, and the lemma is then proven. Numerical means floating point: the result is evidence, not proof, and the statement says "numerically" and is not called proven. The proof names the software and version when the original gives them. When the text does not say which kind of computation was used, insert `[[EVIDENCE: exact or floating-point computation, and the software used]]`. Adding such a lemma is the one exception to the last item of Rule 6. The report lists each new lemma, and `compare` lists it as a changed formal environment.
- **Engineering detail to delete.** Remove detail about running the code that no reported result depends on:
  - regularization or jitter values added for numerical stability that are not part of the model;

  - guards, safeguards, fallbacks, and retries that never triggered;
  - random-number arrangements, such as which seed went to which run or how streams were split;
  - solver tolerances, iteration caps, and time limits;
  - verification residuals and other self-checks of the computation ("the residual is below 10^{-12}");
  - names of scripts, functions, files, directories, and environments;
  - accounts of debugging, reruns, and failed runs.

- **Reproduction information to keep.** Keep every value a reported result depends on: model parameters and hyperparameters, sample sizes, the number of replications or runs, data sources and versions, the algorithm and its stopping rule or tolerance when a reported gap, error, or runtime depends on it, the software and version behind a computer-assisted proof, the hardware when runtimes are reported, and the availability statement for code and data. A value that might belong to the model stays, and the report asks the author about it. The report suggests moving deleted detail to the code repository or a reproducibility appendix, and never moves it there itself.
- **Deleted numbers.** Every number removed under this rule appears in the report as an intended change, with its original line. `compare` lists it under numbers removed.

12. Structure audit, reported and changed only on request

Use the script's structure output and a full reading. Report:

- a section with exactly one subsection, to merge into its section or to pair with a second subsection;
- sections that could merge;
- the number of main-text results, intermediate results that could become appendix lemmas, and results that restate another under a different parameter and could become corollaries or remarks;
- paragraphs with more than one point, and places where the argument jumps;
- ordinal headings in non-manuscript documents (4.6) when cleanup mode fixes the structure.

Venue register

- Conference papers with strict page limits, common in computer science and engineering: short and results-first, contributions stated as specific results, limitations stated plainly and once.
- Journal papers: the research question grounded in a concrete setting, the literature positioned by stream with the specific difference from the closest work, assumptions stated with their justification, and each main result followed by its interpretation and the condition under which it holds. Understatement of tone, full statement of results.
- Referee response letters: one reply per comment, quoting the comment and stating the change and its location. A disagreement gives the argument and the evidence. One round of thanks at the top.
- Grant proposals (NSF, NIH, and similar): read `references/proposals.md`.
- Chinese text, including textbooks, theses, and reports: read `references/chinese.md`.
- Earlier papers by the author: see "Style reference first".

Final checklist

- Every change corresponds to a numbered Rule, and changes without one are reverted.

- In cleanup mode, every sentence without a rule match is verbatim, and matched sentences changed only in the matched part.
- No new names, numbers, dates, quotations, sources, causal claims, or examples. Every content word traces to the original or the supplied material.
- No real qualifier, concession, claim, or conclusion was dropped. Each qualifier that `compare` lists as lost was removed under Rule 7 and appears in the report.
- Headings, section order, paragraph order, lists, tables, and quotations are in place.
- No invented contrast, cue colon, list lead-in, restatement sentence, spoken opener, meta-narrative sentence, preview, or recap remains outside the conclusion section.
- No adjacent sentences share one skeleton, and no paragraph pads its point with parallel cases.
- No dash used as punctuation, no paragraph crowded with semicolons or asides, and no figurative or persona label outside established terms.
- Every coined label is an established or defined term, has been replaced by what it does, or appears in the report with a suggestion, and the revision adds none.
- Summary phrases with a specific value nearby now carry that value, and nominalized verbs are restored.
- Empirical verbs match their evidence, proven results carry their conditions, and established results read as established, without apology or defence.
- Methods and proofs describe the method and the argument, not the running of code. Computer-assisted steps are lemmas that say whether the computation is exact or numerical, deleted engineering detail is in the report with its numbers, and every value a reported result depends on remains.
- Every missing piece of evidence is a placeholder in polish mode or a report item in cleanup mode, and any remaining placeholder is announced at the top of the report as blocking submission.
- None of the “Not reasons to change” features was edited for its own sake.
- `compare` reports numbers, citation keys, labels, math, and formal statements unchanged, or lists each difference as intended.

Output

1. The complete revised text. A `.tex` or other file input comes back as the full revised file, never as a list of edit descriptions.
2. A short report, written under the same Rules:
   - when the revised text contains any `[[EVIDENCE: ...]]` placeholder, a first line, in the language of the text, telling the author that the text is not ready to submit: each placeholder must be replaced with a value or source from their own data, or its claim deleted, before the text goes to a journal, conference, or reviewer;
   - the mode and, if used, the style reference;
   - changes grouped by Rule, with counts;
   - claims scoped, strengthened under Rule 7, or given evidence pointers;
   - every placeholder the author must fill, and in cleanup mode every claim that needs evidence;

- engineering detail deleted under Rule 11, each item with its numbers and original line, values kept because they might belong to the model, and computer-assisted steps restated as lemmas;
- structural issues (Rule 12), figurative names chosen by the author, and coined labels kept without a definition, each with a suggested fix (Rule 2);
- the final `compare` result for numbers, citation keys, labels, math, formal statements, and qualifiers.

3

Worked examples

Each example states the context the author supplied and the mode, which is polish unless stated. Every number, result, and reference in an “After” text comes from that context. Where the context lacks what a sentence needs, the “After” text carries a placeholder instead of a plausible value. Citation keys such as ref1 and ref2 are placeholders.

The running context for every example except Example 7: the paper estimates the size of a closed animal population from repeated capture surveys in which some animals are easier to catch than others. The stratified estimator divides the animals into strata by catchability, estimates the size of each stratum separately, and adds the estimates. Theorem 2 proves that the stratified estimator is consistent as the number of sampling rounds grows, for a closed population with equal catchability within each of at least two strata. Table 3 reports, over 1,200 simulated populations, a mean relative error of 0.8% (standard error 0.1%) for the stratified estimator and 2.9% for the best unstratified estimator. Its parameter grid varies the capture probability and the number of strata. No statistical test is reported, and the text does not state the largest error over the grid.

1. Introduction opening (Rules 1, 5, 6)

Before:

> In recent years, wildlife population monitoring has attracted increasing attention from both academia and practitioners. With the rapid development of sensing technologies, ecologists face intricate trade-offs between survey cost and precision, making the choice of estimator of paramount importance. In this paper, we propose a novel framework that sheds light on this pivotal problem and offers rich practical insights. Extensive numerical experiments demonstrate the effectiveness of our approach.

After:

> Capture surveys estimate the size of an animal population from repeated captures, and the estimate is biased when some animals are easier to catch than others. We study this problem for a closed population in which catchability is equal within each of at least two strata. We prove that the stratified estimator is consistent as the number of sampling rounds grows (Theorem 2). The estimator estimates the size of each stratum separately and adds the estimates. Across 1,200 simulated populations its mean relative error is 0.8% (standard error 0.1%), compared with 2.9% for the best unstratified estimator (Table 3).

What changed: the opening states the estimation problem instead of its popularity. “Novel”, “sheds light”, “pivotal”, “rich practical insights”, and “extensive” go, and “demonstrate the effectiveness” becomes the numbers from Table 3 with their comparator. “Prove” stays because Theorem 2 is proven in the paper, and the sentence carries the theorem’s conditions.

2. Scope of a proven result (Rule 6)

Before:

> We prove that the stratified estimator is optimal, which provides a simple and robust solution for practitioners.

After:

> We prove that the stratified estimator is consistent as the number of sampling rounds grows, for a closed population with equal catchability within each of at least two strata (Theorem 2).

What changed: “prove” stays. “Optimal” becomes “consistent”, which is what Theorem 2 states, with the theorem’s conditions. “Robust” goes because the paper contains no robustness result, and “for practitioners” goes because no part of the paper concerns field use. The report offers a literal version of the simplicity point: the estimator needs only the stratum labels and the capture counts.

3. Numerical claims and placeholders (Rule 5)

Before:

> Extensive numerical experiments demonstrate that our estimator significantly outperforms existing benchmarks and remains robust across a wide range of settings.

After:

> Across the 1,200 simulated populations in Table 3, the mean relative error of the stratified estimator is 0.8% (standard error 0.1%), compared with 2.9% for the best unstratified estimator. Over the grid of capture probabilities and stratum counts in Table 3, the error does not exceed [[EVIDENCE: largest error over the grid in Table 3]].

What changed: “significantly” goes because no test is reported. The comparison is with the strongest benchmark in Table 3 and uses the statistic the table reports. “Robust across a wide range of settings” becomes the actual grid plus a placeholder for the one number the text does not supply. A plausible value such as “1.5%” in that place would be a fabrication.

4. Splitting a long sentence without fragmenting it (Rule 8)

Before:

> Existing population estimators, though effective in specific sampling regimes, lack theoretical guarantees underpinning their performance, which restricts their applicability and hinders further development.

Too fragmented, never the target:

> Existing estimators work in some regimes. They lack guarantees. That restricts applicability.

After:

> Existing population estimators perform well in specific sampling regimes, but they have no performance guarantee. Without a guarantee, it is unclear in which surveys an estimator will perform well, and this uncertainty limits both its application and its further development.

What changed: each sentence carries one step, and the logical relations stay explicit (“but”, “Without a guarantee”). The fragmented version has short sentences and drops the relations, which leaves the reader to reconstruct the argument.

5. Practical implication (Rules 1, 6)

Additional context: Proposition 4 shows that, under the conditions of Theorem 2, the number of sampling rounds needed to reach a target precision decreases in the capture probability and increases in the difference in catchability across strata.

Before:

> These results offer rich practical insights: practitioners should carefully balance survey effort and precision when designing monitoring programs.

After:

> The number of sampling rounds needed for a given precision decreases as the capture probability rises and increases with the difference in catchability across strata (Proposition 4). A survey whose strata differ more in catchability should therefore plan more rounds even when the capture probability does not change.

What changed: the implication is the comparative static of Proposition 4, with its direction and its source. The second sentence follows from the second half of the proposition and claims nothing beyond it.

6. Citation list (Rule 10)

Before:

> Population estimation has been widely studied \citep{ref1, ref2, ref3, ref4, ref5}.

After:

> Population estimation has a large literature \citep{ref1, ref2, ref3, ref4, ref5}.

Report entry: “Line 14 cites five works without saying how any of them differs from this paper. Consider naming the closest one and the difference.”

What changed: all five keys stay, and no cited work is described beyond what the text already says.

7. Referee response (venue register)

Context: the authors added Section 5.2, which allows animals to move between strata. Proposition 6 extends Theorem 2 to this case, with the proof in Appendix C.3.

Before:

> We sincerely thank the reviewer for this insightful and valuable comment. Following the reviewer's excellent suggestion, we have carefully revised the manuscript accordingly to address this important point.

After:

> We added Section 5.2, which allows animals to move between strata. Proposition 6 extends Theorem 2 to this case, and its proof is in Appendix C.3.

What changed: the reply states the change and its location. Thanks appear once, at the top of the letter.

8. Invented contrast, cue colon, and restatement: polish versus cleanup (Rules 4, 6)

Before:

> The difficulty of estimating population size is not computational; it lies in unequal catchability. The key insight is simple: under the conditions of Theorem 2, stratifying by catchability loses little. This means that the estimator does not need a model of individual behavior.

After, polish mode:

> The difficulty of estimating population size lies in unequal catchability. Under the conditions of Theorem 2, the stratified estimator is consistent as the number of sampling rounds grows, so it does not need a model of individual behavior.

After, cleanup mode:

> The difficulty of estimating population size lies in unequal catchability. Under the conditions of Theorem 2, stratifying by catchability loses little, so the estimator does not need a model of individual behavior.

What changed: no one in the text claims the difficulty is computational, so the contrast goes (4.1) and the author's own claim stays. "The key insight is simple:" announces the sentence instead of stating it (4.2). The "This means that" sentence restates the previous one and merges into it with "so" (4.4). Polish mode also replaces "loses little" with the statement of Theorem 2 under Rule 6. Cleanup mode leaves the claim's strength alone, so "loses little" stays and the report says: "Line 3: 'loses little' is weaker and vaguer than Theorem 2. Consider stating the theorem's result."

9. Qualifiers survive the edit (edit discipline, Rules 7 and 10)

Context: Table 3 varies the capture probability, and the text reports no test of performance at low capture probabilities.

Before:

> Our results suggest that the stratified estimator may perform well when the capture probability is low. It is worth noting that this is a pivotal finding.

Wrong:

> Our results show that the stratified estimator performs well when the capture probability is low.

After:

> Our results suggest that the stratified estimator may perform well when the capture probability is low.

What changed: the second sentence carries only announced emphasis and a significance word (Rule 1) and goes, since it adds no claim. “Suggest” and “may” stay because the text has no direct evidence for low capture probabilities, so the uncertainty is real. The wrong version gains confidence by changing the claim, and `compare` lists both lost qualifiers. Example 10 shows the opposite case, where the evidence is in the text and the hedge goes.

10. Stance and meta-narrative (Rules 4.9, 7)

Before:

> In this section, we present our main numerical results. Although our approach is only a preliminary attempt and has several limitations, the results may potentially suggest that the stratified estimator could perform better than existing estimators: its mean relative error is 0.8% (standard error 0.1%), compared with 2.9% for the best unstratified estimator (Table 3). To summarize, the stratified estimator performs well.

After:

> Across 1,200 simulated populations, the stratified estimator has a mean relative error of 0.8% (standard error 0.1%), compared with 2.9% for the best unstratified estimator (Table 3).

What changed: the opening sentence describes the section instead of stating a result, and the closing sentence recaps it (4.9). “Only a preliminary attempt” and “has several limitations” apologize without naming a limitation that bounds this result (Rule 7). “May potentially suggest … could perform better” stacks three hedges on numbers the paper itself reports, so the result reads as a result (Rule 7). The population count comes from Table 3 in the running context. `compare` lists “may” and “suggest” as lost qualifiers and the new number 1,200, and the report records each as intended.

11. Coined labels (Rules 2, 6)

Context: Section 4.2 states that each heuristic solution is checked for feasibility and that its objective is recomputed with the exact solver. The text says nothing more about the "robustness layer", and no proposition covers it.

Before:

> Each heuristic solution then passes through our rigorous certification protocol, and a robustness layer filters the remaining candidates, ensuring trustworthy decisions.

After:

> We then check each heuristic solution for feasibility and recompute its objective with the exact solver (Section 4.2). A robustness layer filters the remaining candidates.

Report item: "robustness layer" appears once and is never defined. State the filtering rule it applies, or name the step plainly ("a second filter"). "Robust" asserts a property that no result in the text supports (Rule 6).

What changed: the text states what the "certification protocol" does, so the label becomes those two steps, and "rigorous" goes with it (Rule 1). "Certification" also promised a guarantee the procedure does not give (Rule 6). The text never says what the "robustness layer" does, so the label stays and the report asks the author to define it. Inventing a filtering rule would add content. "Ensuring trustworthy decisions" is a trailing evaluation (Rule 1) and goes. The section number comes from the running context, so the report records the new number 4.2 that `compare` lists as intended. `audit` lists both labels under "no definition or citation", and `compare` on a revision that added a label such as "validation pipeline" would list it as new.

12. A proof written as a work log (Rule 11)

Context: the proof of Theorem 2 needs $g(x) \ge 0.37$ on $[0,1]$, where g is defined in (12). The text does not say whether the bound was computed exactly or in floating point. Table 2 reports estimation errors averaged over 200 replications with $n = 500$, and no reported number depends on the solver settings.

Before:

> Our verification pipeline proceeds in three stages. A sanity check confirms that $g(x) > 0$ on $[0,1]$, the residual was below 10^{-12}, and the certified bound $g(x) \ge 0.37$ holds. We ran `verify_bound.py` with seed 42 for the first run and seed 7 for the second. A jitter of 10^{-8} was added to the diagonal, and the safeguard against singular matrices never triggered. The solver tolerance was 10^{-9} with an iteration cap of 5000. We report the mean over 200 replications with sample size $n = 500$ (Table 2).

After:

> **Lemma 4.** For all $x \in [0,1]$, $g(x) \ge 0.37$.
> *Proof.* We evaluate g on $[0,1]$ by [[EVIDENCE: exact (interval or rational arithmetic) or floating-point computation, and the software used]].
> We report the mean over 200 replications with sample size $n = 500$ (Table 2).

Report items:

- Not ready to submit: the placeholder in the proof of Lemma 4 must say how g was evaluated. With exact arithmetic, Lemma 4 is proven. With floating point, it becomes a numerical statement, and Theorem 2 then rests on numerical evidence, which its statement must say.
- Deleted under Rule 11 as intended changes: the residual bound 10^{-12} (second sentence), the seeds 42 and 7 and the script name `verify_bound.py` (third), the jitter 10^{-8} and the safeguard that never triggered (fourth), and the solver tolerance 10^{-9} and iteration cap 5000 (fifth). These belong in the code repository or a reproducibility appendix if the venue asks for them.
- Kept for reproduction: 200 replications and $n = 500$, since Table 2 depends on both.
- Lemma 4 is new. Its statement is the bound the original text asserts. `compare` lists its number and its math as added, and in a `.tex` file it also lists the new lemma environment.

What changed: “verification pipeline”, “sanity check”, and “certified bound” framed the proof as a run of code (Rule 11), so the bound becomes a lemma with a proof. “Certified” goes because nothing in the text shows an exact computation. “$g(x) > 0$” goes because Lemma 4 implies it. The seeds, jitter, safeguard, tolerance, iteration cap, residual, and script name describe how the code ran, and no reported result depends on them, so each is deleted and listed with its number.

4

Grant proposals

The evidence and citation rules of SKILL.md apply to proposals unchanged. Agency requirements change between cycles, so the current solicitation and the agency's instructions take precedence over any structural statement below: for NSF, the PAPPG and its supplements, and for NIH, the funding opportunity and the application instructions. For other agencies, follow the agency's template and review criteria, and Sections 3 to 5 still apply.

How a proposal differs from a paper

A proposal asks reviewers to fund work that has not been done, so its central claims concern a plan: what the team will attempt, why the attempt can succeed, and what the result would change. The claim and evidence rules of a paper become claim and feasibility rules (Section 5). Statements of a long-term goal and of expected impact, which Rule 1 removes from a paper, are expected in a proposal when the plan and the preliminary evidence support them. The audit script still lists words such as "transformative" and "pioneer"; keep each one only where its support appears in the same paragraph.

1. Structure

Reviewers usually form a view of a proposal from its opening pages and read the rest to check that view, so the opening pages deserve the most editing.

- **NSF.** The one-page Project Summary has three parts, Overview, Intellectual Merit, and Broader Impacts, each introduced by its heading on a line of its own and each understandable without the others. Within its first one or two pages, the Project Description should give the long-term goal, the goal of this proposal, the gap, the thrusts or aims, and the expected results, preferably with one overview figure. Broader Impacts should name concrete activities that are connected to the research.
- **NIH (R01 and other research project grants).** The Specific Aims page summarizes the proposal on one page and is the page most reviewers read closely. A common order runs from the problem, what is known, and the critical gap, through the long-term goal and the central hypothesis with its rationale, and the objective of this application, to two or three aims and what changes if they succeed. Each aim is a one-line goal, a phrase on the approach, and the expected outcome. The Research Strategy then has Significance, Innovation, and Approach sections. For due dates on or after January 25, 2025, most research project grants are reviewed under the simplified framework: Factor 1 (Importance of the Research, covering Significance and Innovation) and Factor 2 (Rigor and Feasibility, covering the Approach) are scored from 1 to 9, and Factor 3 (Expertise and Resources, covering Investigators and Environment) is rated only as appropriate or as having gaps.

2. What the opening pages must contain

By the end of page 1 (NIH Specific Aims) or pages 2 to 3 (NSF Project Description), the reader should know five things: why the problem matters, stated concretely, what is missing and what the gap costs, the central idea of the approach in one sentence, the aims or thrusts in parallel form, and the expected results. If any of these is missing or appears late, fix that before editing later sections.

3. Weak patterns to fix

- **Vague importance.** Watch for "this is an important and timely problem" and "X has many applications". State the specific consequence of the gap.

Context supplied by the author: the proposal derives bounds that link measurement noise in wearable sensors to diagnostic error, and current diagnostic models that use such data are tuned by trial and error.

Before: Understanding this problem is critically important.

After: Diagnostic models that use wearable-sensor data are tuned by trial and error, because no available bound links measurement noise to diagnostic error. This proposal derives such bounds.

- **Method as aim.** An aim that names a technique instead of a question or an outcome.

Before: Aim 2: Apply deep learning to camera-trap images.

After: Aim 2: Determine for which species and image conditions an automated classifier identifies individual animals with a lower error rate than trained human coders.

- **Dependent aims.** If Aims 2 and 3 cannot proceed when Aim 1 fails, reviewers treat the plan as fragile. State the aims so that each produces a result on its own; where one aim depends on another, state the fallback.
- **Ambition without feasibility.** Each ambitious claim needs support placed next to it: preliminary data, a prior result or publication, a classical result the work builds on, or a collaborator's commitment. Write the support into the same sentence or the next one ("the preliminary result in Figure 2 shows …").
- **Generic Broader Impacts or training plans.** Watch for "we will mentor students and disseminate the results through talks and papers". Replace with the specific programs, courses, software, or datasets that the plan contains, connected to the research, with counts where the plan gives them.
- **Hedged central hypothesis.** The hypothesis on the Specific Aims page is a falsifiable commitment ("We hypothesize that …"), not "we will explore whether … may …". Hedging belongs in the Approach, where expected results and alternatives are interpreted.

4. Keep or add

- A long-term goal stated at the start, with this proposal as a defined step toward it.
- Run-in headings that make the text easy to scan: **Goal.**, **Motivation.**, **Innovation.**, *Aim N (one-line statement).*
- One concrete running example used across all aims.
- Each aim's central question stated as a question.
- The connection to named classical results that the work extends (a known inequality, policy class, or test), when the connection is real.
- The team's record as feasibility evidence, placed early: prior funded work, preliminary results, publications, collaborators, and partners. Use only the record the author supplied. Never invent funding, results, partners, or letters.

5. Claim and feasibility

For every aim and every promised outcome, check whether the ambition is matched by a stated means: preliminary data, a prior method, a classical foundation, a collaborator, or a staged plan with a fallback. If it is, keep the ambitious verb. If it is not, attach the missing support or narrow the claim to what the plan supports. When the support does not exist in the supplied material, insert `[[EVIDENCE: what is needed]]` and list it in the report; do not write around the gap.

5

A checker run

A results paragraph in LaTeX, its revision, and the output of both checker commands. The revision removes the preview sentence, the recap, the self-deprecation, the invented contrast, and a stack of three hedges on numbers the paper reports in its own table. `compare` confirms that the numbers and the table reference are unchanged and lists the three removed qualifiers, which the report records as intended stance edits under Rule 7. Exit status 0 means that no invariant changed. Conservation warnings leave the exit status unchanged.

Original

> \section{Results}
> In this section, we present the main results of our simulation study. Although our approach is only a preliminary attempt, the results may potentially suggest that the stratified estimator reduces bias: its mean relative error is 0.8\% (standard error 0.1\%), compared with 2.9\% for the best unstratified estimator (Table~\ref{tab:sim}). The difficulty is not computational; it lies in unequal catchability. To summarize, stratifying by catchability removes most of the bias.

Revision

> \section{Results}
> The stratified estimator has a mean relative error of 0.8\% (standard error 0.1\%), compared with 2.9\% for the best unstratified estimator (Table~\ref{tab:sim}). The difficulty lies in unequal catchability, and stratifying by catchability removes most of the bias.

Output of `audit orig.tex` (excerpt)

```
file: pdf-src/demo/en/orig.tex  (LaTeX)

== AI-tell candidates (orig.tex) ==
  contrast                      1   (invented contrasts (SKILL.md Rule 4.1); keep when the rejected position is named or cited)
      line 2: ...fied estimator (Table [ref]). The difficulty >>is not computational; it lies<< in unequal catchability. To
summarize, strat...
  meta                          2   (meta-narrative, previews, and recaps (SKILL.md Rule 4.9); the conclusion section keeps its
summary)
      line 2: >>In this section, we present<< the main results of our simulation study. Al...
      line 2: ...putational; it lies in unequal catchability. >>To summarize<<, stratifying by catchability removes most of...
  defensive                     3   (self-deprecation, defensive phrasing, and stacked hedges (SKILL.md Rule 7))
      line 2: ...ation study. Although our approach is only a >>preliminary attempt<<, the results may potentially suggest that
th...
      line 2: ...r simulation study. Although our approach is >>only a preliminary attempt<<, the results may potentially suggest
that th...
      line 2: ...h is only a preliminary attempt, the results >>may potentially<< suggest that the stratified estimator reduce...
```

Output of `compare orig.tex rev.tex --changes` (excerpt)

```
original: pdf-src/demo/en/orig.tex
revised:  pdf-src/demo/en/rev.tex

== Invariants (must be unchanged unless the change is intended and reported) ==
```

```
  numbers                          unchanged (3)
  citations                        unchanged (0)
  cross-references and labels      unchanged (1)
  math                             unchanged (0 segments)
  formal statements                unchanged (0)

  RESULT: all invariants hold

== Conservation warnings (do not affect the exit status) ==
  qualifiers and concessions: 3 -> 0
    LOST (each one strengthens a claim unless it was intended):
      may x1  original line 2: ...only a preliminary attempt, the results >>may<< potentially suggest that the stratified...
      potentially x1  original line 2: ... a preliminary attempt, the results may >>potentially<< suggest that the stratified
estimator r...
      suggest x1  original line 2: ...ry attempt, the results may potentially >>suggest<< that the stratified estimator reduces
b...
  capitalized terms not in the original: 0
  quoted strings not in the original: 0
  sentences: 1 of 5 unchanged; 0 changed, 2 merged or split, 0 added, 0 removed

== Changed sentences (match each to a rule in the report) ==
  C1 [merged] original line 2
    - In this section, we present the main results of our simulation study.
    - Although our approach is only a preliminary attempt, the results may potentially suggest that the stratified estimator
reduces bias: its mean relative error is 0.8% (standard error 0.1%), compared with 2.9% for the best unstratified estimator (Table
[ref]).
    + The stratified estimator has a mean relative error of 0.8% (standard error 0.1%), compared with 2.9% for the best
unstratified estimator (Table [ref]).
  C2 [merged] original line 2
    - The difficulty is not computational; it lies in unequal catchability.
    - To summarize, stratifying by catchability removes most of the bias.
    + The difficulty lies in unequal catchability, and stratifying by catchability removes most of the bias.

  1 conservation warning group(s): check that each listed item is intended
```

6

Lexicon categories

The audit reads its candidates from `references/lexicon.txt`, one regular expression per line. Categories marked *rewrite* usually need a change. Categories marked *review* are correct in some contexts, such as "prove" for a proven result or "navigation" in a robotics paper, and need a decision. Add phrases or silence terms that are technical in your field by editing the file.

Category	Action	What it flags	Patterns
`ai_vocab`	rewrite	vocabulary typical of LLM output	32
`hype`	rewrite	significance claims without content	18
`opener`	rewrite	formulaic openers	7
`intensifier`	rewrite	size words without numbers	9
`novelty`	rewrite	novelty padding	5
`emphasis`	rewrite	announced emphasis	5
`connective`	rewrite	additive connectives (a problem when they open consecutive sentences)	1
`figurative`	rewrite	figurative language; state the mechanism instead	25
`figurative_or_technical`	review	figurative in general prose, technical in some fields; keep only the technical use	17
`copula`	rewrite	copula avoidance	2
`parallelism`	rewrite	negative parallelism or contrast used for emphasis	2
`ing_tail`	rewrite	-ing clause adding a vague evaluation	1
`vague_hedge`	rewrite	vague hedges; quantify or cut	5
`implications`	rewrite	boilerplate about insights and implications	9
`response`	rewrite	referee-response boilerplate	4
`scope`	review	claim verbs and quantifiers; check each against the evidence or the proven statement	16
`contrast`	rewrite	invented contrasts (SKILL.md Rule 4.1); keep when the rejected position is named or cited	5
`cue_colon`	rewrite	cue phrase plus colon (SKILL.md Rule 4.2)	5
`spoken_opener`	rewrite	spoken openers (SKILL.md Rule 1)	1
`persona`	rewrite	persona metaphors (SKILL.md Rule 2)	2
`restate`	review	restatement sentences (SKILL.md Rule 4.4); keep only when they add a conclusion	2
`meta`	rewrite	meta-narrative, previews, and recaps (SKILL.md Rule 4.9); the conclusion section keeps its summary	9

Category	Action	What it flags	Patterns
`defensive`	rewrite	self-deprecation, defensive phrasing, and stacked hedges (SKILL.md Rule 7)	14
`worklog`	review	work-log framing and engineering detail in methods and proofs (SKILL.md Rule 11); keep values a reported result depends on	17
`zh_boilerplate`	rewrite	中文套话与空泛表述	12
`zh_figurative`	rewrite	比喻或流行语，改用字面表述	19
`zh_figurative_or_technical`	review	一般行文中是比喻，在部分领域是术语；只保留术语用法	7
`zh_emphasis`	rewrite	空洞强调与连接词堆叠	10
`zh_verbose`	rewrite	冗余动词结构，改用直接动词	4
`zh_scope`	review	需要核对证据或命题范围的断言词	7
`zh_contrast`	rewrite	翻案腔（chinese.md 规则 11.1）；被推翻的观点确有出处时保留	7
`zh_cue_colon`	rewrite	提示语加冒号（chinese.md 规则 11.2）	1
`zh_spoken_opener`	rewrite	口语起手式（chinese.md 规则 1）	1
`zh_persona`	rewrite	拟人化喻体（chinese.md 规则 2）	1
`zh_translationese`	review	翻译腔（chinese.md 规则 6、11.4）；只处理这几种，表示数学条件的"当……时"保留	5
`zh_meta`	rewrite	元叙事、预告与小结（chinese.md 规则 11.8）；全文结论部分保留总结	6
`zh_defensive`	rewrite	自贬、防御与叠用限定（chinese.md 规则 13）	6
`zh_worklog`	review	方法与证明里的工作日志式写法和工程细节（chinese.md 规则 14）；报告结果所依赖的取值保留	14
`label_head`	label check	head nouns of coined labels such as "certification protocol"	21
`label_ok`	label check	established terms the coined-label check skips	11
`zh_label_head`	label check	拼凑标签的中心词，如"认证协议""保障机制"	13
`zh_label_ok`	label check	有定义的术语，拼凑标签检查跳过	6

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